

Project

Comparative Study of Signal Peptide Prediction Methods Based on Statistical Models and Machine Learning Models

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Abstract

Motivation: Signal peptides (Signal Peptide, SP) are key sequence features in the protein secretion pathway. Accurate prediction of signal peptides is of great significance for protein localization annotation and functional studies. Traditional signal peptide prediction methods mostly rely on sequence features based on empirical rules or statistical models, while in recent years machine learning methods have shown stronger modeling capabilities in this task. However, under the premise of strictly controlling data quality and confounding factors, the differences among methods in prediction performance and error patterns still require systematic evaluation.

Results: In this study, we constructed a high-quality, low-redundancy eukaryotic protein signal peptide dataset based on UniProtKB, ensuring that the only systematic difference between positive and negative samples was whether a signal peptide was present. We implemented and compared the classical von Heijne statistical model and a machine learning model based on Support Vector Machine (SVM). On an independent benchmark test set, we conducted a systematic evaluation of the two methods from two levels: overall performance metrics and false positive and false negative patterns. The results show that the SVM model outperforms the von Heijne model in overall prediction performance and false positive control, while the latter still has advantages in recognizing typical signal peptide sequence features and biological interpretability.

Keywords: signal peptide prediction; von Heijne model; SVM; machine learning; protein localization

1. Introduction

Signal peptides (Signal Peptide, SP) are short amino acid sequences located at the N-terminus of proteins and are usually responsible for directing newly synthesized polypeptides to the endoplasmic reticulum membrane and into the classical secretory pathway. This process is a key step in protein secretion, membrane localization, and organelle targeting in eukaryotic cells, and has a decisive impact on protein subcellular localization, maturation processing, and functional execution ^[1-2]. Therefore, accurate identification of signal peptides is an important fundamental problem in protein function annotation and secretory protein research ^[3].

Early signal peptide prediction methods were mainly based on statistical analysis of known signal peptide sequences and empirical rule modeling. Among them, the most representative is the position-specific statistical model proposed by von Heijne, which characterizes typical signal peptide features such as the hydrophobic core region (H region) and cleavage site sequence preferences by analyzing the distribution characteristics of amino acids around the signal peptide cleavage site ^[4]. Methods developed based on similar ideas have been widely adopted in early signal peptide prediction tools and have had an important impact due to their good interpretability ^[5]. However, when facing non-typical signal peptides or complex sequence backgrounds, the predictive performance of such methods may be limited.

With the development of machine learning methods, signal peptide prediction strategies based on feature engineering and classification models have gradually attracted attention. Supervised learning methods such as Support Vector Machine (SVM) integrate multiple sequence features and to some extent overcome the limitations of single statistical models, showing good generalization ability in multiple biological sequence classification tasks [6-7]. However, performance comparisons among different methods are often influenced by dataset construction strategies, redundancy control, and differences in biological background, making it difficult to directly compare conclusions.

In this study, we take signal peptide prediction as the research object and, under the premise of strictly controlling data sources, redundancy levels, and biological background consistency, systematically compare the performance of the classical statistical model (von Heijne) and the machine learning - based SVM method on this task. In addition to overall predictive performance, we further analyze the sequence characteristics of misclassified samples, with particular attention to the impact of biological factors such as transmembrane helices on prediction results, thereby revealing the applicable scenarios and potential limitations of different methods.

2. Materials and Methods

2.1 Data sources and initial filtering

In this study, the UniProtKB database was used as the primary source of protein sequences. All sequences were derived from manually reviewed (reviewed) entries and met the following basic criteria: non-fragment proteins (fragment = false), presence of protein-level evidence (existence = 1), and sequence length not less than 40 amino acids.

To avoid species bias, all data were restricted to eukaryotic organisms (Eukaryota), without imposing additional constraints on specific species, thereby ensuring biological background diversity of the dataset.

Table 1. Definition and filtering criteria of positive and negative samples

This table summarizes the unified filtering criteria applied to positive samples (with signal peptides) and negative samples (without signal peptides) in terms of data source, species range, annotation quality, signal peptide characteristics, and redundancy removal.

Item	Positive samples	Negative samples
Data source	UniProtKB	UniProtKB
Species range	Eukaryota	Eukaryota
Manually reviewed	reviewed = true	reviewed = true
Protein evidence level	Protein existence = 1	Protein existence = 1
Fragment	fragment = false	fragment = false
Signal peptide annotation	Experimentally validated signal peptide	Experimentally validated signal peptide
Signal peptide length	≥ 14 aa	/
Subcellular localization	Secretory-related pathways	Cytosol / Nucleus / Mitochondria, etc.
Redundancy removal	MMseqs2	MMseqs2
Protein sequence length	≥ 40 aa	≥ 40 aa

2.2 Definition of positive and negative samples

Positive sample definition: Eukaryotic protein sequences containing signal peptides, where the signal peptide cleavage site has experimental evidence and the signal peptide length is not less than 14 amino acids, in order to ensure the integrity of typical signal peptide structures.

Negative sample definition: Eukaryotic protein sequences that do not contain signal peptides, whose subcellular localization has been experimentally verified to be non-secretory-related compartments, including cytosol, nucleus, mitochondria, endoplasmic reticulum, peroxisome, and cell membrane.

Under the above definitions, the only systematic difference between positive and negative samples is whether a signal peptide is present, thereby minimizing potential confounding factors to the greatest extent.

2.3 Redundancy removal and data splitting

To reduce the impact of highly similar sequences on model training and evaluation, FASTA sequences of positive and negative samples were clustered separately using MMseqs2.

- a) Clustering parameters were set to a sequence identity threshold of 30% and a coverage of 40%, and only the representative sequence of each cluster was retained.
- b) Based on the clustering results, entries corresponding to representative sequences were mapped back to the original annotation table, thereby generating redundancy-reduced positive and negative sample datasets.
- c) The redundancy-reduced dataset was randomly split into a training set and a benchmark test set at a ratio of 80% and 20%. In addition, 5-fold cross-validation was constructed on the training set for model selection and parameter optimization.

2.4 von Heijne statistical model

The von Heijne model constructs a position-specific scoring matrix (Position-Specific Weight Matrix, PSWM) based on sequence windows around signal peptide cleavage sites.

- a) During the training phase, only positive sample sequences were used. Fixed-length sequence windows were extracted according to known cleavage sites, and the occurrence frequencies of amino acids at each position were calculated. Pseudocounts were introduced simultaneously to avoid zero-probability issues.
- b) During the prediction phase, sliding window scanning was performed on the N-terminal region of the query sequence, scores were calculated for each window, and the maximum score was taken as the signal peptide prediction score for that sequence. The score was converted into a binary classification result using a predefined threshold.

2.5 Support Vector Machine – based model

The SVM model formulates the signal peptide prediction problem as a binary classification task. Positive samples were labeled as 1 and negative samples as 0. Multiple features were extracted for each sequence, including N-terminal amino acid composition ratios and hydrophobicity-related features, and assembled into numerical feature vectors. Model parameters were selected through 5-fold cross-validation on the training set. The model was then retrained on the full training set using the optimal parameters and applied to the benchmark test set for evaluation.

2.6 Performance evaluation metrics

Model performance was evaluated using multiple metrics, including accuracy (Accuracy), sensitivity (Sensitivity / Recall), precision (Precision / PPV), and the Matthews correlation coefficient (MCC). In addition, confusion matrices were constructed to visually display the distribution of prediction results.

3. Dataset Characterization

Before model training and performance evaluation, systematic statistical analyses were performed on the constructed dataset to assess its compositional characteristics, biological plausibility, and consistency between the training set and the benchmark test set. This section provides a comprehensive analysis of the positive sample dataset from the perspectives of species distribution, amino acid composition features, and sequence length distribution.

3.1 Species and taxonomic distribution of positive samples

First, the evolutionary source distributions of positive samples (signal peptide – containing sequences) in the training set and the benchmark test set were statistically analyzed. The results show that, both in the training set and in the benchmark test set, positive samples exhibit highly consistent distribution patterns at the taxonomic level.

At the kingdom level, positive samples are mainly derived from Metazoa, accounting for more than 80% of the total, followed by Plants and Fungi, while other eukaryotic lineages contribute relatively small proportions (Figure 1). The training set and the benchmark test set show highly similar proportions in kingdom-level distributions, indicating that the dataset split did not introduce significant taxonomic bias.

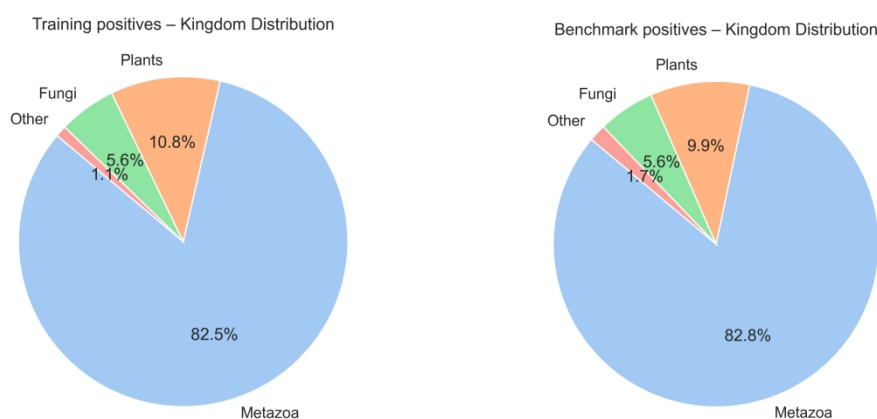


Figure 1. Kingdom distribution of positive samples in the training set and the benchmark test set

This figure shows the proportion of signal peptide – positive samples in the major eukaryotic groups (Metazoa, Plants, Fungi, and Others) in the training and benchmark test sets. The two distributions are highly consistent, indicating that the dataset split did not introduce obvious species bias.

Further analysis of the species-level distribution of positive samples shows that model organisms and commonly studied species such as *Homo sapiens*, *Mus musculus*, *Rattus norvegicus*, and *Bos taurus* account for the major proportion. At the same time, the dataset also contains a large number of sequences from other eukaryotic species (Figures 2 – 3). This distribution pattern reflects, to some extent, the research bias of manually reviewed UniProt entries, while still ensuring a certain level of species diversity in the training data.

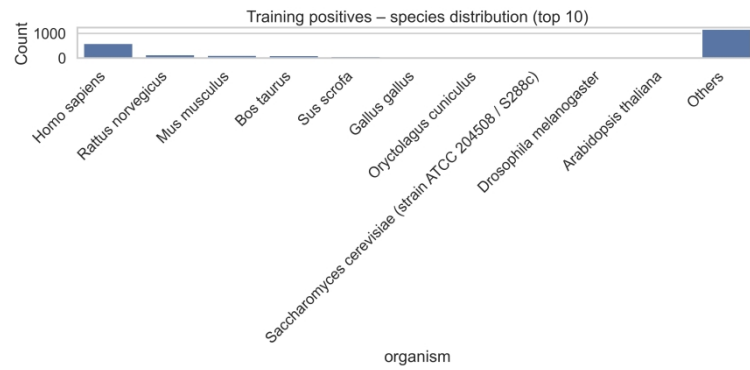


Figure 2. Species distribution of positive samples in the training set (Top 10)

The bar chart shows the top ten species ranked by the number of signal peptide – positive samples in the training set. The remaining species are grouped as “Others”, reflecting the annotation concentration of model organisms in the UniProt

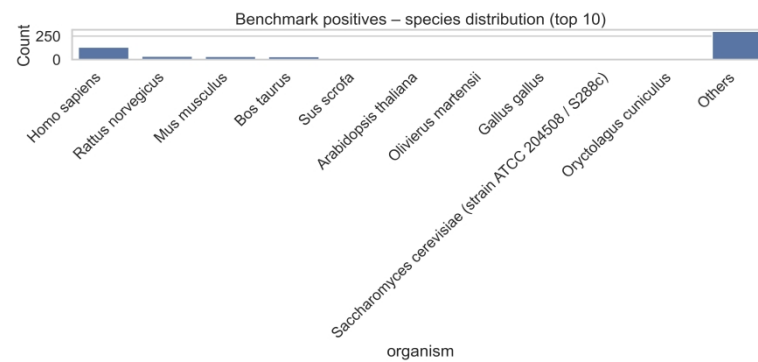


Figure 3. Species distribution of positive samples in the benchmark test set (Top 10)

Compared with the training set, the benchmark test set shows a highly consistent composition in terms of major species, further indicating that the dataset splitting process did not introduce significant species bias.

3.2 Amino acid composition analysis of signal peptides

To further verify whether signal peptide sequences in the positive samples conform to known biological characteristics, the amino acid composition of signal peptide regions in the training set was statistically analyzed and compared with the overall amino acid distribution of all protein sequences in the SwissProt database.

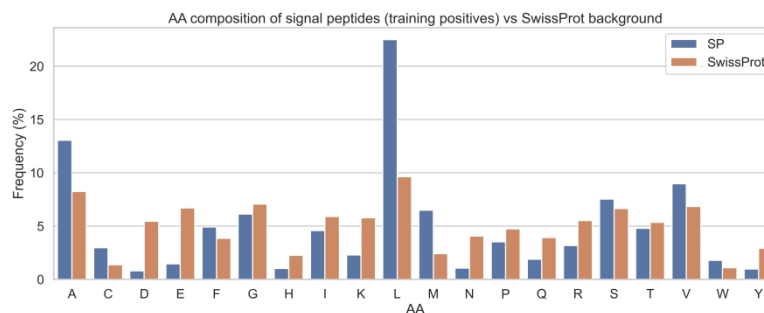


Figure 4. Comparison of amino acid composition between signal peptide regions and SwissProt background sequences

The figure compares the composition proportions of 20 amino acids between signal peptide regions in the training set and the SwissProt global protein background.

Signal peptide regions show a significant enrichment of hydrophobic amino acids (such as Leu, Val, and Ala), while charged amino acids are relatively depleted, which is consistent with the classical biological characteristics of signal peptides.

The results show that the proportions of hydrophobic amino acids (such as Leu, Val, Ala, and Ile) in signal peptide regions are significantly higher than those in the SwissProt background distribution, while the proportions of charged or polar amino acids (such as Asp, Glu, and Lys) are markedly reduced (Figure 4). This phenomenon is highly consistent with the biological characteristic of a highly hydrophobic H region in classical signal peptide structures, indicating that the constructed positive sample dataset exhibits good biological plausibility at the level of sequence composition.

3.3 Signal peptide length distribution characteristics

A statistical analysis was performed on the length distribution of signal peptides in the positive samples. The results show that, both in the training set and in the benchmark test set, signal peptide lengths are mainly concentrated between 15 and 30 amino acids, with a peak distribution around 18 – 22 amino acids (Figure 5).

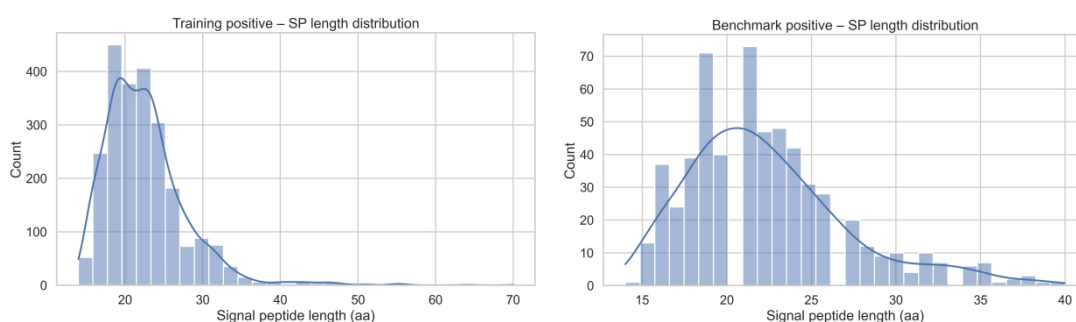


Figure 5. Signal peptide length distribution in the training set and the benchmark test set

The signal peptide length distribution in the benchmark test set is highly consistent with that in the training set, indicating that signal peptide length characteristics remain stable before and after dataset splitting.

This distribution pattern is highly consistent with classical descriptions of eukaryotic signal peptide lengths in the literature, further validating the rationality of the positive sample definition and filtering strategy.

3.4 Distribution analysis of full-length protein lengths in positive and negative samples

In addition to signal peptides themselves, further comparisons were made between positive and negative samples in terms of the length distribution of full-length protein sequences. Boxplot analysis shows that there are certain differences in overall protein length distributions between positive and negative samples: negative samples contain more ultra-long protein sequences, and the number of outliers is markedly higher than that of positive samples (Figure 6).

This phenomenon is consistently observed in both the training set and the benchmark test set, indicating that the observed difference is not introduced by dataset splitting, but rather reflects the objective length distribution differences between secretory and non-secretory proteins in real biological data.

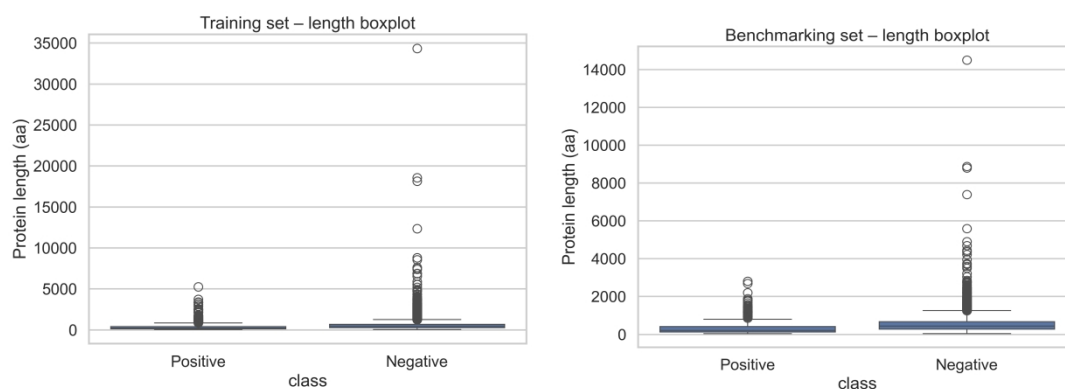


Figure 6. Full-length protein length distribution

The boxplots compare the differences in full-length protein length distributions between positive and negative samples. Negative samples tend to include longer protein sequences overall, while positive samples show a more concentrated distribution. This trend is consistent across both the training set and the benchmark test set.

3.5 Summary

In summary, through systematic analyses of taxonomic distribution, species composition, amino acid composition, and sequence length characteristics, it can be confirmed that the dataset constructed in this study is highly consistent with known biological characteristics of signal peptides. Moreover, the training set and the benchmark test set maintain good consistency across multiple feature dimensions. These results provide a reliable data foundation for subsequent model training, performance evaluation, and result interpretation.

4. Experimental Results

4.1 Five-fold cross-validation and threshold determination for the von Heijne model

To evaluate the stability of the von Heijne statistical model on the training set and to determine an appropriate classification threshold, five-fold cross-validation was performed on the redundancy-reduced training dataset. The optimal threshold values ranged from 5.77 to 6.32, with an average threshold of 6.056. The variation among folds was small, indicating that the statistical model is not sensitive to the specific data split and exhibits good stability. The corresponding classification performance remained consistent across different folds, with accuracy ranging from 0.923 to 0.942 and Matthews correlation coefficient (MCC) ranging from 0.680 to 0.742. No obvious performance anomalies were observed. These results indicate that the von Heijne model shows a stable learning process on the training set and that threshold selection has strong robustness.

4.2 Overall prediction performance on the independent benchmark test set

Based on the five-fold cross-validation results, the final von Heijne model was retrained using the full training set, and the average optimal threshold obtained from cross-validation (6.056) was applied to predictions on the independent benchmark test set.

On the benchmark test set, the von Heijne model achieved an accuracy of 0.928 and an MCC of 0.692, indicating that the model retains strong discriminative power under class-imbalanced conditions (Figure 7).

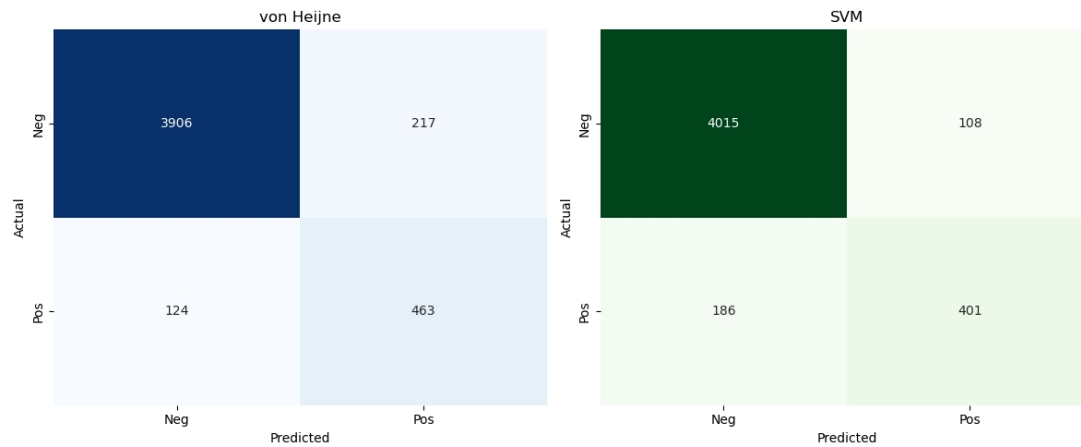


Figure 7. Comparison of confusion matrices between the von Heijne model and the SVM model on the benchmark test set. The left panel shows the von Heijne model, and the right panel shows the SVM model. The SVM model significantly reduces the number of false positives, while the number of false negatives increases slightly, reflecting the differences between the two

methods in terms of error patterns.

From the confusion matrix, it can be observed that most positive and negative samples are correctly classified, although a certain number of misclassifications still exist. Specifically, there are 217 false positive (FP) samples and 124 false negative (FN) samples. Subsequent analyses will further investigate the characteristics of these misclassified samples in detail.

4.3 Analysis of sequence features learned by the von Heijne model

To analyze the sequence features captured by the von Heijne model, the trained PSWM (log-odds weight matrix) was visualized (Figure 8).

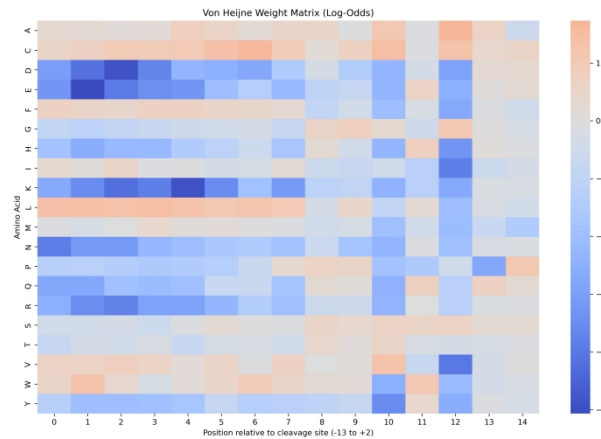


Figure 8. Position-specific weight matrix (log-odds) of the von Heijne model

The heatmap shows the log-odds weight distribution of amino acids at different positions around the cleavage site, reflecting the typical sequence features of the signal peptide N region, H region, and the vicinity of the cleavage site.

This weight matrix clearly reflects the structural characteristics of classical signal peptides. In the N region upstream of the cleavage site, basic amino acids (such as K and R) have relatively high weights; in the H region (approximately positions -13 to -6), hydrophobic amino acids (such as L, I, V, and F) are significantly enriched; near the cleavage site, small amino acids (such as A, S, and G) show a clear preference, consistent with the classical $(-3, -1)$ cleavage rule.

Furthermore, sequence logos were generated for correctly predicted positive samples and false negative samples of the von Heijne model around the signal peptide cleavage site, respectively (Figure 9).

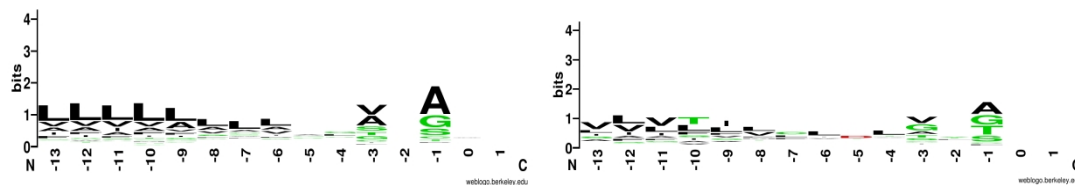


Figure 9. Sequence logo comparison near the signal peptide cleavage site for correctly predicted positive samples (left) and false negative samples of the von Heijne model (right)

在信号肽切割位点附近的序列 logo 对比。

The results show (Figure 9) that although false negative samples still retain certain conserved signals near the cleavage site (positions -3 to -1), the hydrophobicity of the H region (approximately positions -13 to -6) is significantly reduced, lacking the continuous hydrophobic amino acid enrichment required for typical signal peptides. As a result, these sequences receive lower scores in the von Heijne model and are misclassified as non-signal peptide sequences.

Although false negative samples retain certain sequence features near the cleavage site, the reduced hydrophobicity of the H region and the lack of continuous hydrophobic amino acid enrichment typical of signal peptides explain their lower scores in the von Heijne model and their consequent misclassification as non – signal peptide sequences.

4.4 Performance of the SVM model on the benchmark test set

The Support Vector Machine (SVM) – based model formulates the signal peptide prediction problem as a binary classification task and models sequences using multiple feature vectors. The features include N-terminal amino acid composition ratios and hydrophobicity-related indices. Model parameters were optimized through five-fold cross-validation on the training set.

On the independent benchmark test set, the SVM model achieved an accuracy of 0.938 and an MCC of 0.699, showing overall performance slightly superior to that of the von Heijne model. Its confusion matrix is shown in the right panel of Figure 8.

4.5 Analysis of the impact of transmembrane helices on false positive prediction

To further explore the causes of model misclassification, this study statistically analyzed the occurrence of transmembrane helices (Transmembrane Helix, TM helix) in false positive samples.

Among the false positive samples predicted by the von Heijne model, approximately 54.4% contain predicted transmembrane helix structures, whereas this proportion decreases to 40.7% in the SVM model (Figure 10).

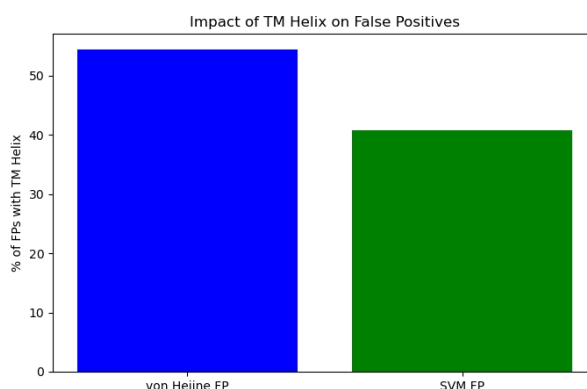


Figure 10. Impact of transmembrane helices (TM helix) on false positive prediction

The bar chart compares the proportion of false positive samples containing transmembrane helices in the von Heijne and SVM models. The results indicate that the SVM model reduces the interference of transmembrane structures on signal peptide prediction to a certain extent.

These results suggest that hydrophobic transmembrane regions are more likely to be confused with the H region of signal peptides at the sequence level, especially in models that rely heavily on local hydrophobicity statistics. In contrast, by integrating global sequence features, the SVM model reduces misclassification caused by transmembrane helices to some extent.

4.6 Overall performance comparison and summary of the two models

Considering all evaluation metrics, both the von Heijne model and the SVM model exhibit strong discriminative ability in the signal peptide prediction task, with MCC values close to 0.7. The von Heijne model has good biological interpretability and performs better in terms of sensitivity, whereas the SVM model shows a clear

advantage in reducing false positives, but correspondingly increases the number of false negatives.

The differences in prediction behavior between the two models reflect the fundamental distinctions between classical statistical methods and machine learning approaches in terms of feature utilization, and also provide directions for subsequent model improvement and integration.

5. Discussion

5.1 Overall model performance and differences

This study compared the performance of the classical von Heijne statistical model and a Support Vector Machine (SVM) – based machine learning model in the signal peptide prediction task. Both models achieved high prediction accuracy and MCC values on the independent benchmark test set, indicating that sequence feature – based methods are effective for this task. However, the two models show clear differences in error patterns, reflecting the fundamental differences between statistical models and discriminative models in terms of feature utilization.

5.2 False positives and interference from transmembrane helices

Analysis of false positive samples indicates that transmembrane helices are an important interfering factor leading to signal peptide misclassification. Because both transmembrane helices and signal peptides exhibit strong hydrophobicity in the N-terminal region, the von Heijne model is more prone to misclassifying them as signal peptides. In contrast, the SVM model integrates multi-dimensional features and reduces the proportion of such false positives to some extent, but cannot completely eliminate the confusion caused by transmembrane regions, indicating inherent limitations of relying solely on sequence-level hydrophobicity information.

5.3 False negatives and non-canonical signal peptides

sequences retain some signal peptide characteristics near the cleavage site, but their hydrophobic core regions are significantly weaker than those of typical signal peptides. This suggests that some true signal peptides may deviate from classical patterns during evolution, making them difficult to accurately identify using models based on fixed position-specific statistical features.

6. Conclusion

This study constructed a high-quality eukaryotic protein sequence dataset and systematically evaluated the performance of statistical models and machine learning models in signal peptide prediction. The results show that the von Heijne model has good biological interpretability, while the SVM model performs better in reducing false positives but is more conservative toward weak signal peptides. The complementary prediction behaviors of the two approaches suggest that integrating biological prior knowledge with machine learning methods has the potential to further improve the accuracy and robustness of signal peptide prediction.

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